MA3521 — Linear ODEs

Name:

Problem 1 $(1+t^2)y' + 2ty = 2t\cos(t^2) \qquad y(0) = 2$

1a. Put the ODE in standard form $y' + p(t)y = q(t)$.

$p(t) =$ $\qquad$ $q(t) =$

1b. Find an integrating factor.

$$\mu(t) = \exp\left(\int \quad dt\right) =$$

1c. Multiply by $\mu(t)$ and integrate.

$$(\mu y)' =$$

$$\mu(t)y(t) = \int \quad dt + C = \quad + C$$

1d. Use the initial condition to evaluate C.

$C =$

Problem 2 $(2+t^3)y' + 3t^2y = 3t^2e^{t^3} \qquad y(0) = 1$

2a. Put the ODE in standard form $y' + p(t)y = q(t)$.

$p(t) =$ $\qquad$ $q(t) =$

2b. Find an integrating factor.

$\mu(t) = \exp\left(\int \quad dt\right) =$

2c. Multiply by $\mu(t)$ and integrate.

$(\mu y)' =$

$\mu(t)y(t) = \int \quad dt + C =$ $\qquad + C$

2d. Use the initial condition to evaluate C.

$C =$